

DOCUMENTATION

SUBMITTED TO: SIR MUHAMMAD SALMAN

SUBMITTED BY:

- AMMAR TANVEER MALIK
- SULEMAN SAEED
- MUHAMMAD USMAN

SECTION: U

SUBJECT: SOFTWARE DESIGN AND ARCHITECTURE

UNIVERSITY OF LAHORE



SOFTWARE DESIGN DOCUMENT (SDD)

YALI.COM

Online Property Management Service

Software Design & Architecture Semester Project

1. Introduction

1.1 Project Overview

Yali.com is a modern web-based Online Property Management Service developed as a semester project for the Software Design & Architecture course. The platform is designed to simplify property management, property searching, and real estate communication through a centralized digital environment.

The name “Yali” originates from the Turkish language and refers to a luxurious mansion or house located along a seashore. Historically, Yali residences symbolize elegance, prestige, and architectural beauty. Inspired by this concept, Yali.com aims to provide users with a premium and trustworthy platform for discovering, listing, and managing properties.

The system serves as a bridge between property owners, property seekers, and administrators. Through an intuitive user interface and a well-structured software architecture, the platform allows users to search for properties, publish listings, book consultations, and manage their real estate activities efficiently.

The project follows modern software engineering principles and utilizes object-oriented design, UML modeling, database design techniques, and layered system architecture to ensure maintainability, scalability, and reliability.

1.2 Purpose of the System

The purpose of Yali.com is to provide a centralized online solution for property management and property discovery. The platform eliminates many challenges associated with traditional property dealings, such as fragmented communication, manual record keeping, and limited accessibility to property information.

The system is designed to:

- Simplify property searching and listing processes.
- Provide a user-friendly environment for buyers, tenants, and property owners.
- Offer efficient communication through consultation booking services.
- Enable administrators to monitor and manage all platform activities.
- Create a secure and transparent ecosystem for real estate management.

1.3 Problem Statement

In traditional real estate systems, property information is often scattered across multiple sources, making it difficult for users to find accurate and updated information. Property owners struggle to market their properties effectively, while buyers and tenants spend significant time searching through unreliable listings.

Additionally, administrative tasks such as managing users, appointments, and property records become increasingly difficult as the number of users grows.

Yali.com addresses these challenges by providing a unified platform where all stakeholders can interact efficiently through a single digital ecosystem.

1.4 Project Objectives

The primary objectives of Yali.com are:

- To provide secure user registration and authentication.
- To facilitate online property listing and management.
- To enable users to search and explore available properties.
- To display complete property details including location, size, price, bedrooms, and bathrooms.
- To provide consultation booking services with company representatives.
- To allow users to manage their own property listings.
- To provide administrative control over users, appointments, listings, and queries.
- To improve transparency and accessibility in property management.
- To build a scalable and maintainable software solution.

2. System Scope

Yali.com provides comprehensive functionality for property management activities.

The system supports three major actors:

Property Seekers

Individuals looking to purchase or rent properties.

Property Owners

Individuals who want to advertise and manage their properties.

Administrators

Authorized personnel responsible for monitoring and controlling system operations.

The platform provides functionality for property searching, property listing, appointment booking, user management, and administrative control.

3. System Architecture

Yali.com follows a Three-Tier Software Architecture consisting of:

Presentation Layer

The presentation layer provides graphical interfaces through which users interact with the system.

Components include:

- User Login Page
- User Signup Page
- Home Page
- Dashboard
- Listings Page
- Property Details Page
- Add Listing Page
- Admin Login Page
- Admin Dashboard

Business Logic Layer

This layer processes user requests and implements all business rules.

Major modules include:

- Authentication Module
- Property Management Module
- Consultation Management Module
- User Management Module
- Administrative Control Module

Data Layer

The data layer manages data storage and retrieval operations.

Stored data includes:

- User Information
- Property Records
- Property Listings
- Consultation Appointments
- User Queries
- Administrative Records

4. System Features

4.1 User Registration and Authentication

The system allows users to create accounts through the Signup page and securely access the platform through the Login page.

Features include:

- New User Registration
- User Authentication
- Session Management
- Secure Access Control

4.2 Home Page

After successful authentication, users are redirected to the Home Page, which serves as the central hub of the platform.

The Home Page displays:

Trending Properties

Users can view featured properties containing:

- Property Location
- Property Price
- Property Size
- Number of Bedrooms
- Number of Bathrooms

Private Consultation

Users can book consultations with company representatives for professional property guidance and assistance.

Add Listing

Property owners can publish their properties by entering details and uploading property images.

User Listings

Users can view all properties they have published through the platform.

Dashboard

Provides access to personal information, records, and user activities.

Listings

Displays all available properties published on the platform.

Footer Section

Contains navigation links, contact information, and platform details.

4.3 Property Listing Management

The Property Listing Module allows users to manage their property advertisements.

Users can:

- Add New Property Listings
- Upload Property Images
- Publish Listings
- Edit Existing Listings
- Delete Listings
- View Published Listings

4.4 Property Exploration

The Listings module enables users to browse available properties.

Functions include:

- Browse Properties
- Search Listings
- View Property Details
- Explore Property Specifications

Selecting “View Details” provides comprehensive information about a property.

4.5 Dashboard Module

The Dashboard provides a personalized environment for each user.

It maintains:

- User Profile Information
- Published Properties
- Property Management Records
- User Activities

4.6 Administrative Module

The Administrative Module provides centralized control over the entire platform.

The Admin Panel allows administrators to manage:

User Management

- View Users
- Update User Information
- Monitor User Activities

Appointment Management

- View Consultation Requests
- Manage Appointments
- Track Consultation Status

Listing Management

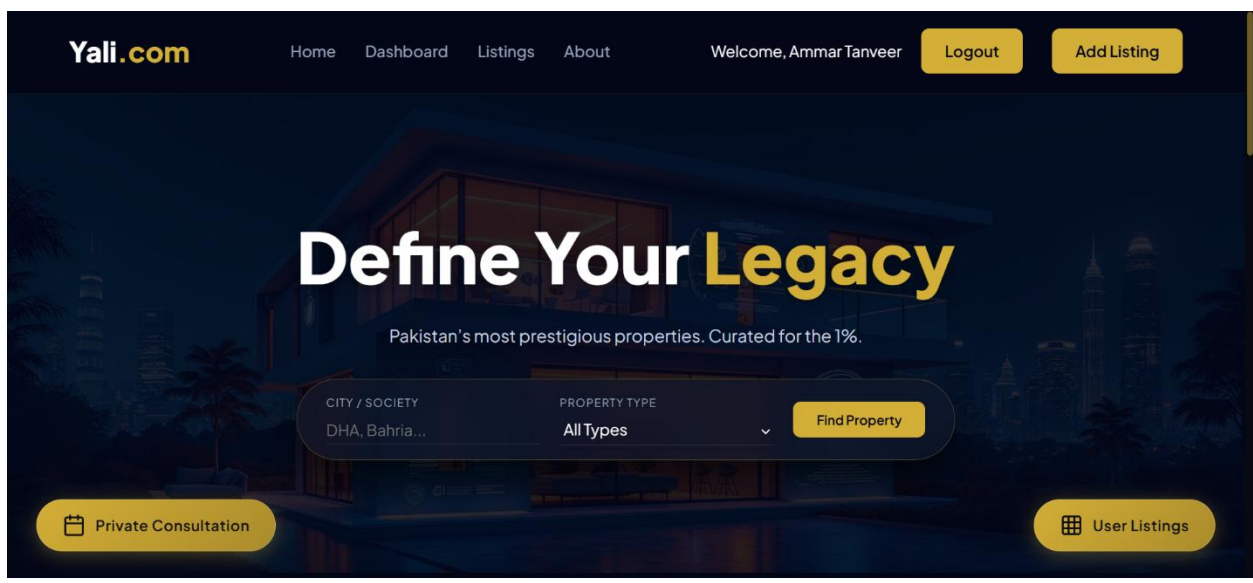
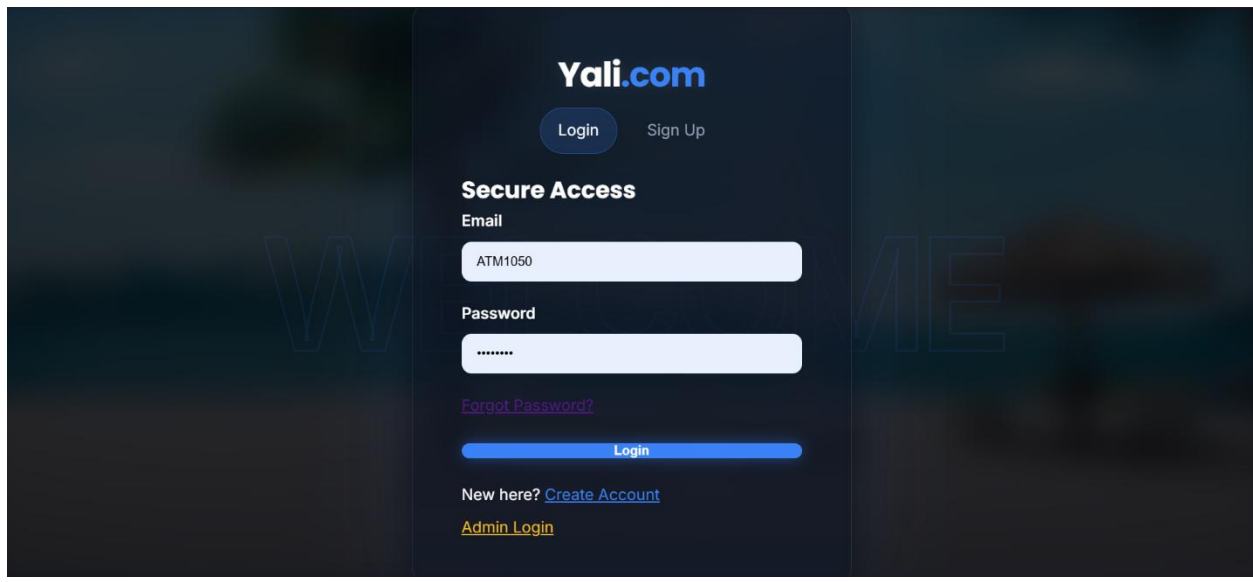
- Monitor Property Listings
- Approve Listings
- Remove Invalid Listings

Query Management

- View User Queries

- Respond to Inquiries
- Monitor User Feedback

User Interface



Yali.com

[Home](#)[Dashboard](#)[Listings](#)[About](#)

Welcome, Ammar Tanveer

Logout

Add Listing

Trending Properties

Hand-picked premium listings in Pakistan's top-tier societies.

DHA Phase 6, Lahore

Classic

Private Consultation

Spanish Estate

Bahria Enclave, Islamabad

1 Kanal • Modern Villa

The Luminary Villa

Gulberg, Lahore

2.5 Kanal • Urban Mansion

The Monolith Mansion

User Listings

Yali.com

[Home](#)[Dashboard](#)[Listings](#)[About](#)

Welcome, Ammar Tanveer

Logout

Add Listing

DHA Phase 8, Lahore

1 Kanal • Contemporary Pavilion

The Glass Pavilion

PKR 10.8 Crore

View Details

Private Consultation

Bani Gala, Islamabad

3 Kanal • Hillside Sanctuary

The Terraced Sanctuary

PKR 22.0 Crore

View Details

DHA Phase 5, Lahore

1.2 Kanal • Executive Residence

The Urban Compact

PKR 13.5 Crore

View Details

User Listings

Yali.com

[Home](#)[Dashboard](#)[Listings](#)[About](#)

Welcome, Ammar Tanveer

Logout

Add Listing

Yali.com

Engineering the future of luxury real estate and exclusive property management across Pakistan and beyond.

Quick Links

[Home](#)[Premium Properties](#)[Our Story](#)

Contact

[Support](#)[List Your Property](#)[Privacy Policy](#)

Private Consultation

© 2026 Yali.com. All rights reserved.

User Listings

localhost/property_project/pages/dashboard.php

Yali.com

Home

Dashboard

Listings

About

Welcome back, **Ammar Tanveer!**

A beautifully polished dashboard for your property portfolio.

ACTIVE LISTINGS

12

↑ 2 this month

TOTAL INQUIRIES

48

↑ 12% vs last week

PENDING VISITS

05

5 meetings scheduled

Yali.com

HomeDashboardListingsAbout

Filters

Reset

LOCATION

All Cities

PRICE RANGE (PKR)

5 Crore50 Crore

PROPERTY TYPE

☒ Modern Villa

☐ Spanish Classic

☐ Penthouse

Apply Filters

DHA Phase 6



PKR 12.5 Crore

Modern Minimalist Estate

1 Kanal • Designer Built • Brand New

View Details

Bahria Town



PKR 8.2 Crore

Classic European Villa

10 Marla • Fully Furnished • Corner

View Details

Yali.com

HomeDashboardListingsAbout

3 Published Listings

Add New Listing

Islamabad



4.5 Crore

Modern Designer House

10 Marla • House / Villa

View Details

Lahore



1.75 Crore

5 Marla Spanish House

5 Marla • House / Villa

View Details

Lahore



122

usma vilaaa

2 marla • House / Villa

View Details

Yali.com

HomeDashboardListingsAbout

Private
Viewing

Curate your lifestyle. Request a confidential consultation or property tour with our elite directors of real estate.

Private Consultation

The Yali Headquarters

FIRST NAME

Enter your first name

LAST NAME

Enter your last name

CONTACT NUMBER

+92 3XX XXXXXXX

EMAIL ADDRESS

name@domain.com

PORTFOLIO SELECTION

Select an exclusive property...

REQUESTED DATE

mm/dd/yyyy

REQUESTED TIME

--:-- --

Yali.com

HomeDashboardListingsAbout

Our Mission

Founded in 2024, Yali.com started with a simple idea: making property transactions as smooth as a single click. We provide a transparent platform for landlords to manage their top-tier assets and for tenants to find their perfect space without the headache of traditional paperwork.

500+

PREMIUM PROPERTIES

10k+

VERIFIED USERS

24/7

CONCIERGE SUPPORT

Yali.com

HomeDashboardListingsAbout

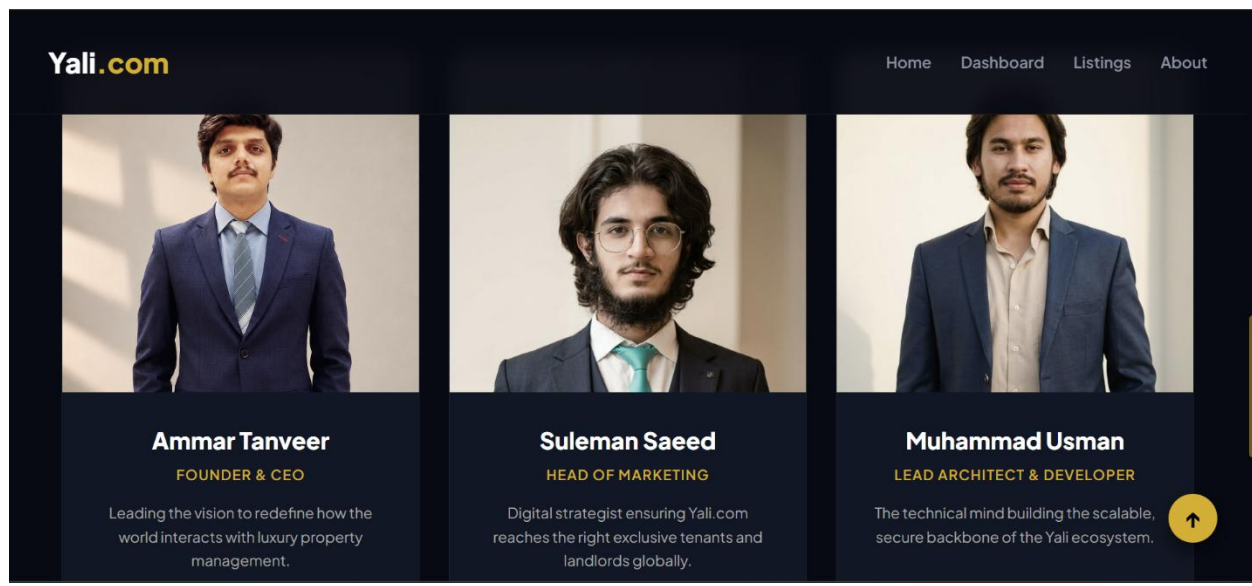
Meet the Visionaries

The core leadership driving innovation in the Yali.com ecosystem.

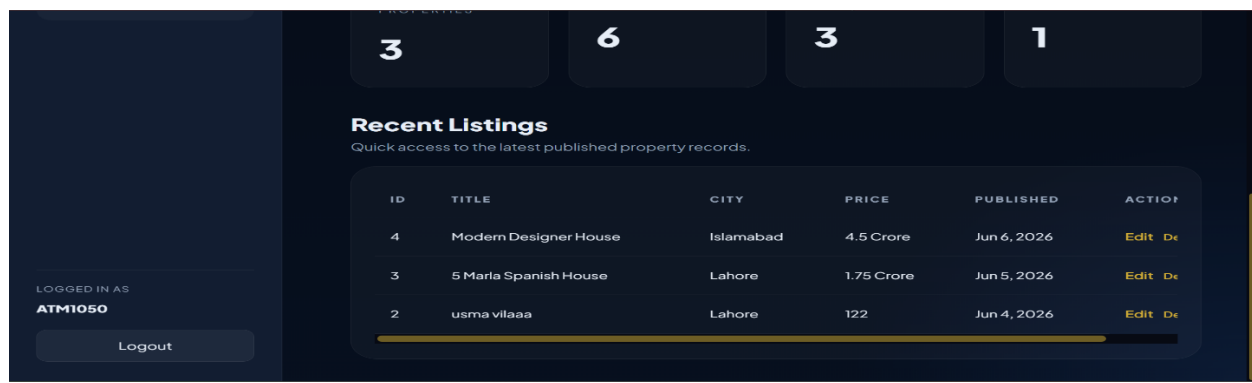
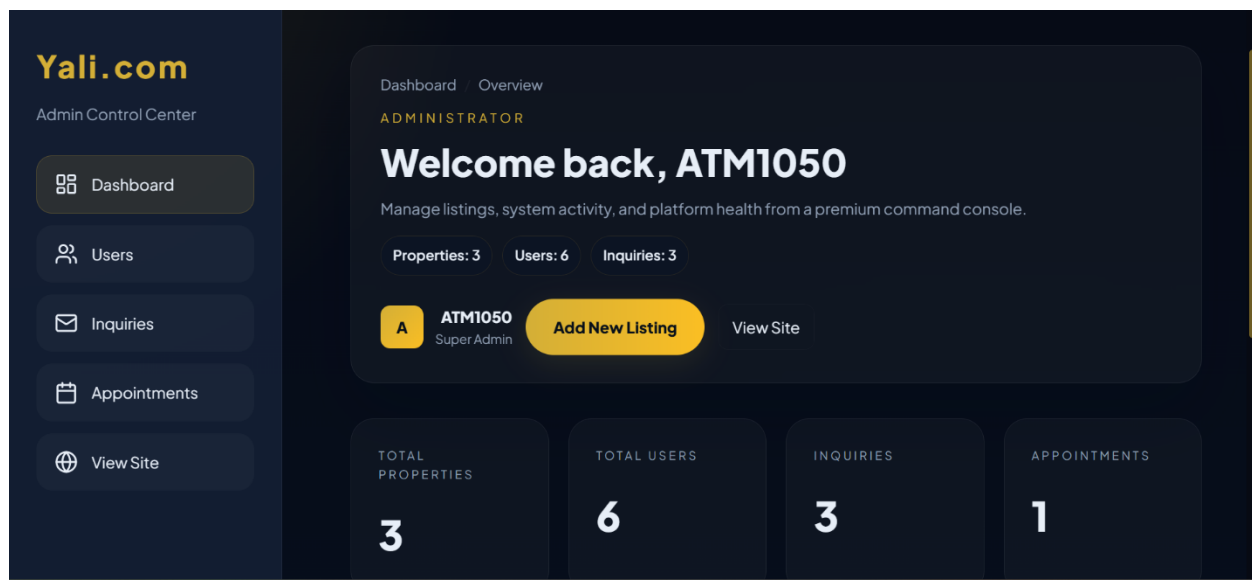








Admin Interface



Yali.com

Admin Control Center

Dashboard

Users

Inquiries

Appointments

View Site

Dashboard / Users

USERS

Manage Registered Users

View and manage user accounts — search, export, or remove as needed.

A

ATM1050Admin

Search users by name or email

Export CSV

ID	NAME	EMAIL	JOINED	A
6	H Habib Ullah habibullah@gmail.com	habibullah@gmail.com	Jun 4, 2026	D
5	U usman usman@gmail.com	usman@gmail.com	Jun 4, 2026	D

Yali.com

Admin Control Center

Dashboard

Users

Inquiries

Appointments

View Site

Dashboard / Inquiries

INQUIRIES

Property Inquiries

Manage incoming messages from prospective buyers and renters.

A

ATM1050Admin

Search inquiries by name, property or phone

Export

ID	PROPERTY	SENDER	PHONE	MESSAGE	RECEIVED	
3	The Luminary Villa	dsfcdsds	+923239897910	vcbdbdfb	Jun 4, 2026	
2	The Terraced Sanctuary	alli	032340404955	street 7 view block a	Jun 4, 2026	
1	The Luminary Villa	Usma	03058537366	123	Jun 4, 2026	

LOGGED IN AS
ATM1050

Yali.com

Admin Control Center

Dashboard

Users

Inquiries

Appointments

View Site

Dashboard / Appointments

APPOINTMENTS

Client Appointments

Review and manage scheduled viewings and meetings with clients.

A

ATM1050Admin

Search by client, property or date

Export

ID	CLIENT	PROPERTY	DATE	TIME	CONTACT
1	usman majeed	The Luminary Villa	2026-05-31	12:22:00	usma123@gmail.com / 03058537366

LOGGED IN AS
ATM1050

5. Functional Requirements

- **FR-01:** The system shall allow users to register, log in, and manage their profiles based on their assigned roles.
- **FR-02:** The system shall enable property owners to add, update, and remove their individual property listings.
- **FR-03:** The system shall provide a search bar and filters for users to query properties based on location, price range, and category.
- **FR-04:** The system shall allow property seekers to submit viewing requests or inquiries directly to property owners.
- **FR-05:** The system shall require admin approval for newly registered property listings before they become visible to the public.
- **FR-06:** The system shall allow users to add properties to a "Favorites" or "Wishlist" for easy access later.
- **FR-07:** The system shall send automated notifications (email or in-app) to users when their property is approved by the admin or when they receive a new inquiry.
- **FR-08:** The system shall provide a "Forgot Password" mechanism allowing users to securely reset their passwords via email verification.
- **FR-09:** The system shall allow property seekers to leave a review and a rating (out of 5 stars) on a property after a verified transaction or visit.
- **FR-10:** The system shall include a "Report Listing" feature, enabling users to flag suspicious, fake, or already sold properties for admin review.

6. Non-Functional Requirements

- Performance
- Security
- Reliability
- Scalability
- Availability
- Usability
- Maintainability

7. Database Design

The database structure is represented through the Entity Relationship Diagram (ERD).

Core entities include:

- User
- Property
- Listing
- Appointment

- Query
- Admin

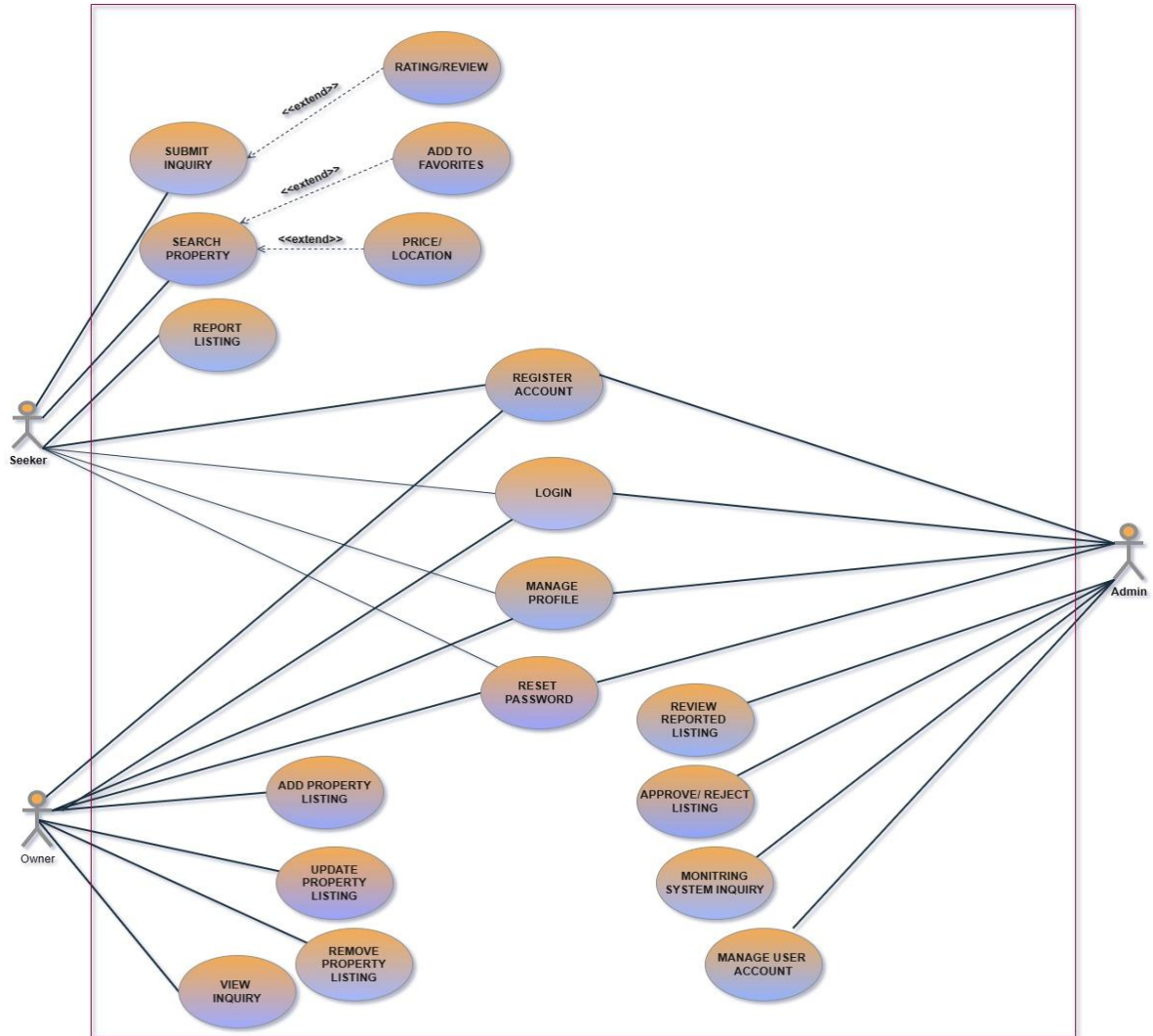
The relationships ensure data consistency, integrity, and efficient retrieval.

8. UML and System Modeling

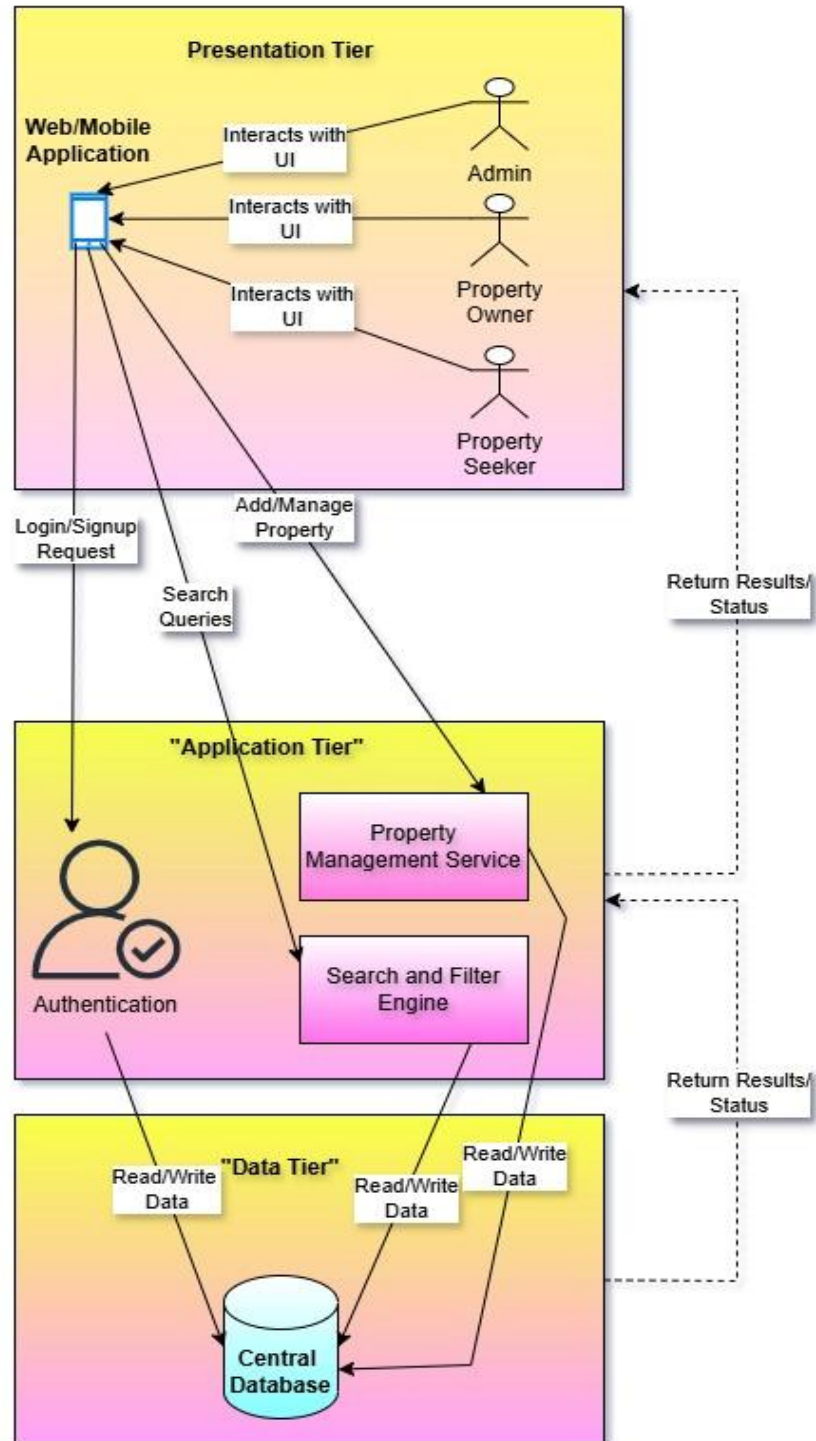
The following diagrams were designed and implemented during system analysis and design:

- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- State Transition Diagram
- Conceptual Class Diagram
- Class Diagram
- Entity Relationship Diagram (ERD)
- DFD Level 0
- DFD Level 1

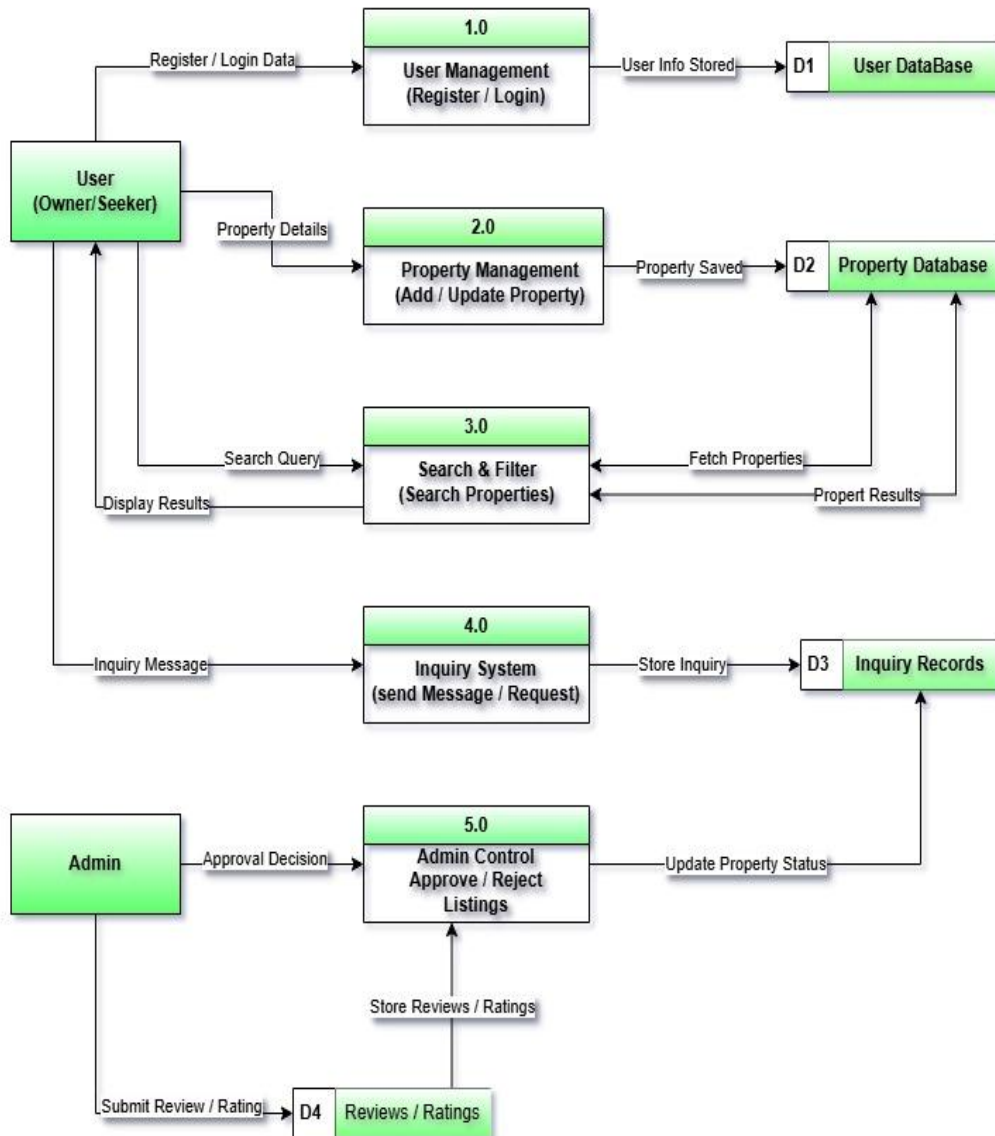
Use Case Diagram:



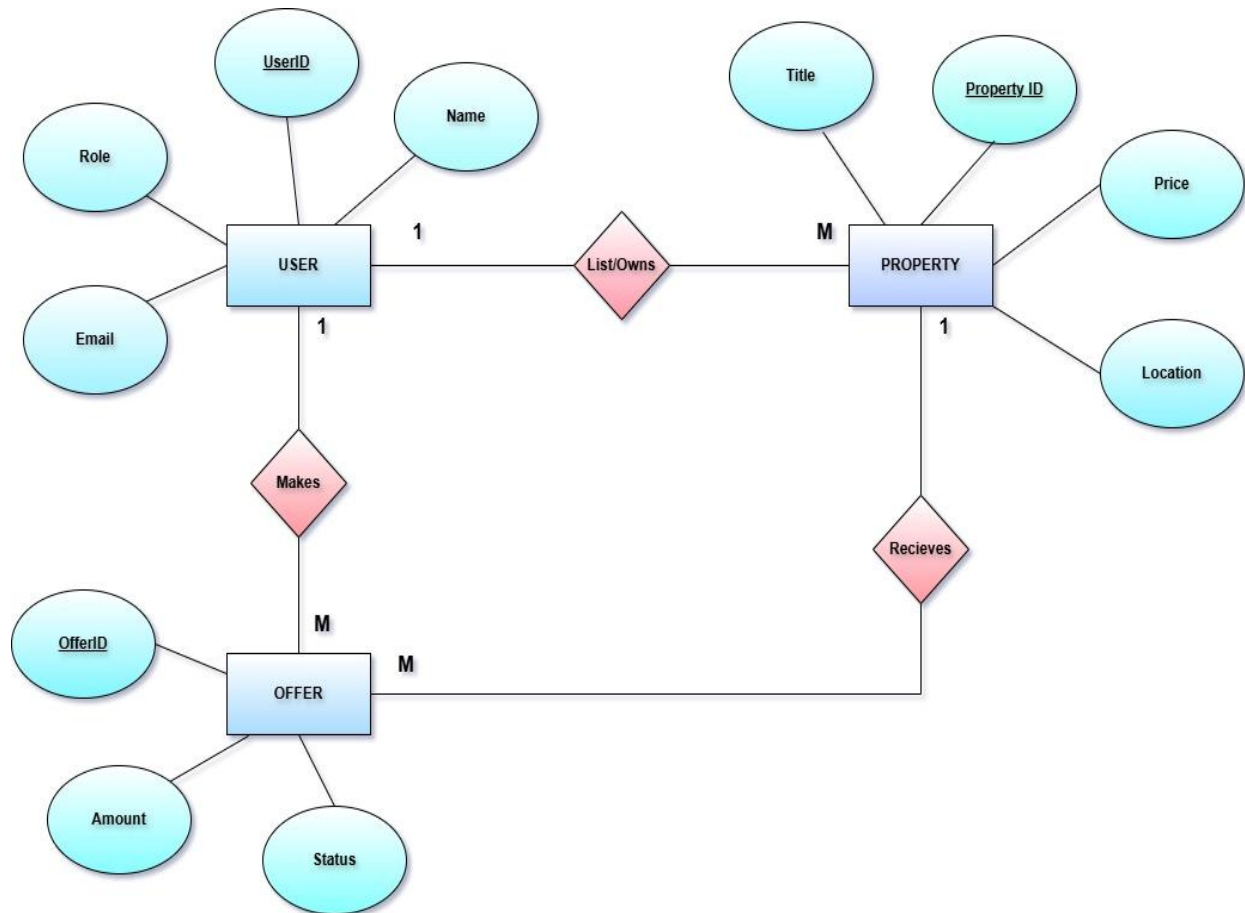
High Level System Flow Diagram



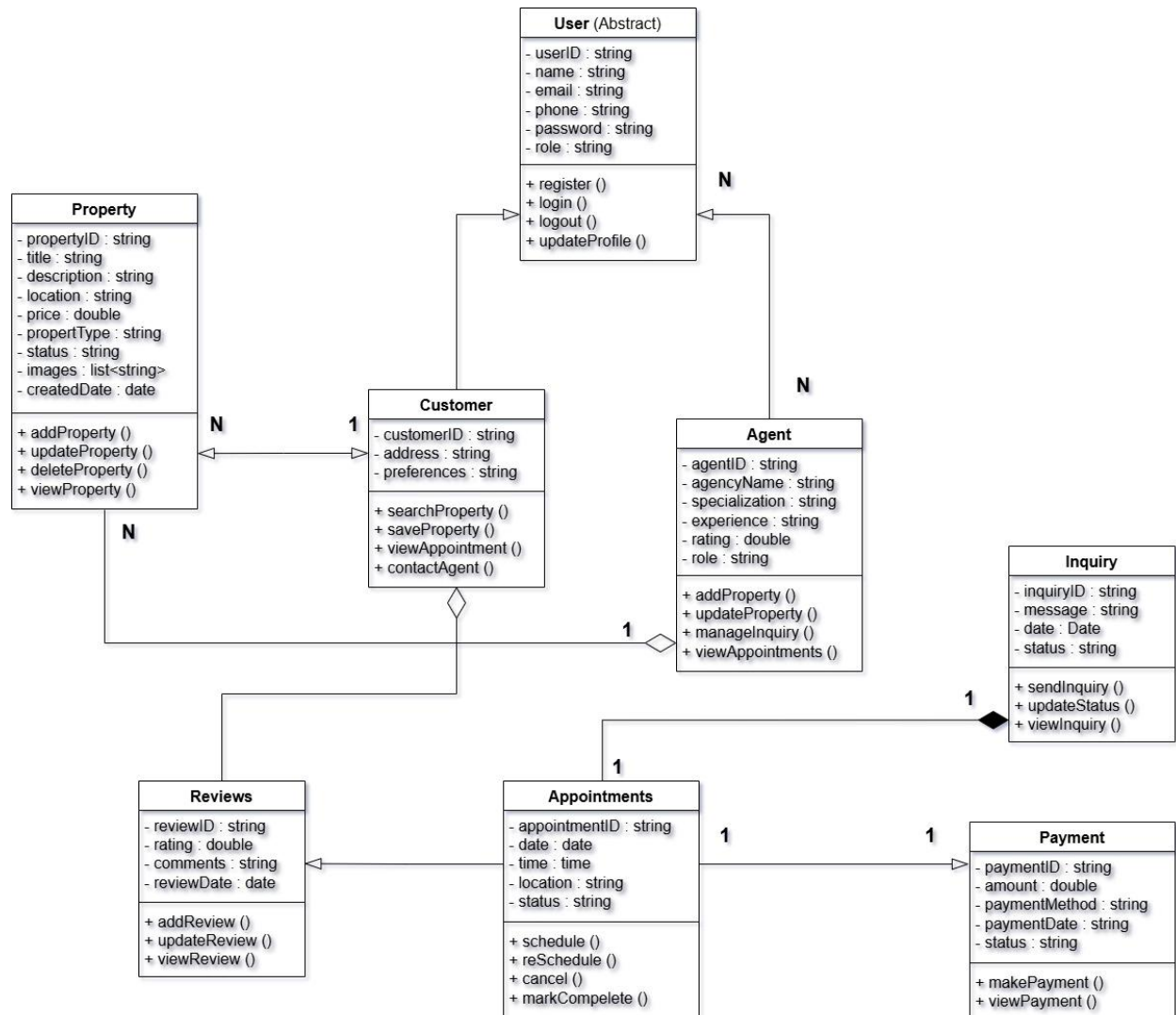
DFD Level 1:



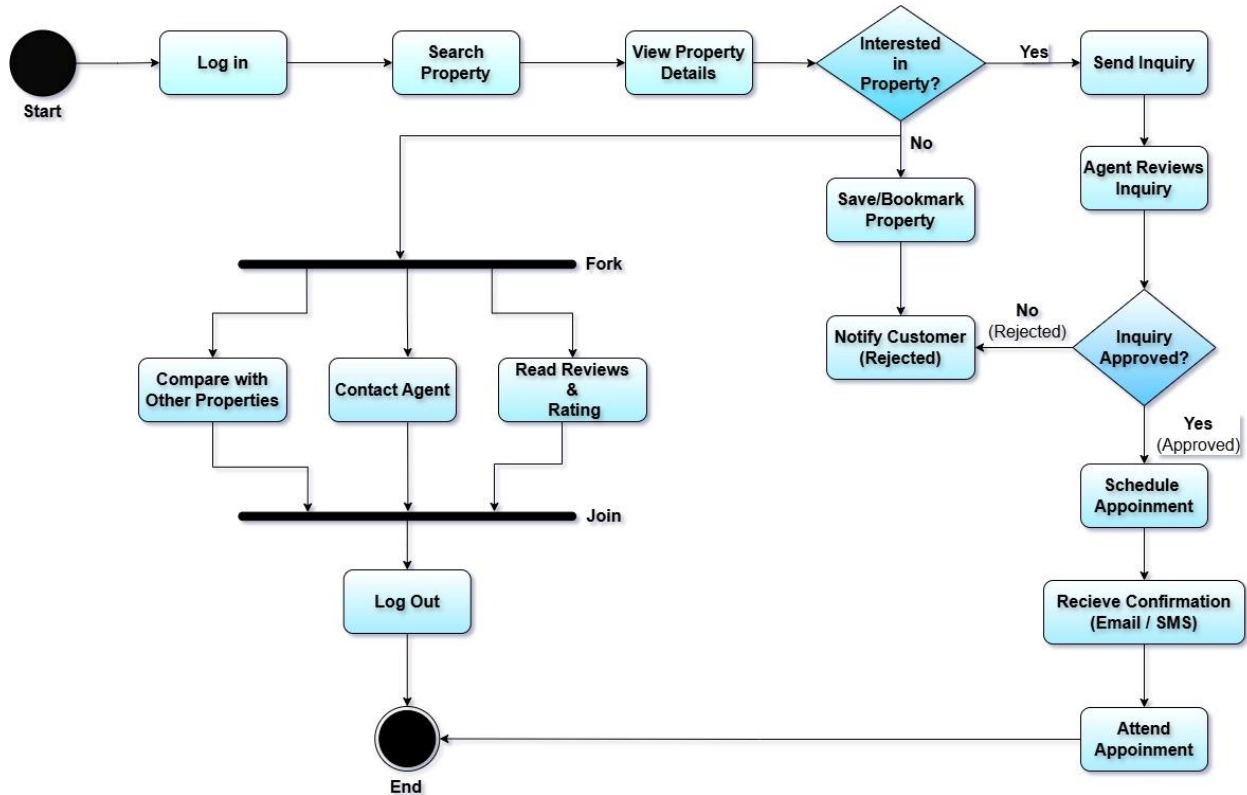
ER DIAGRAM



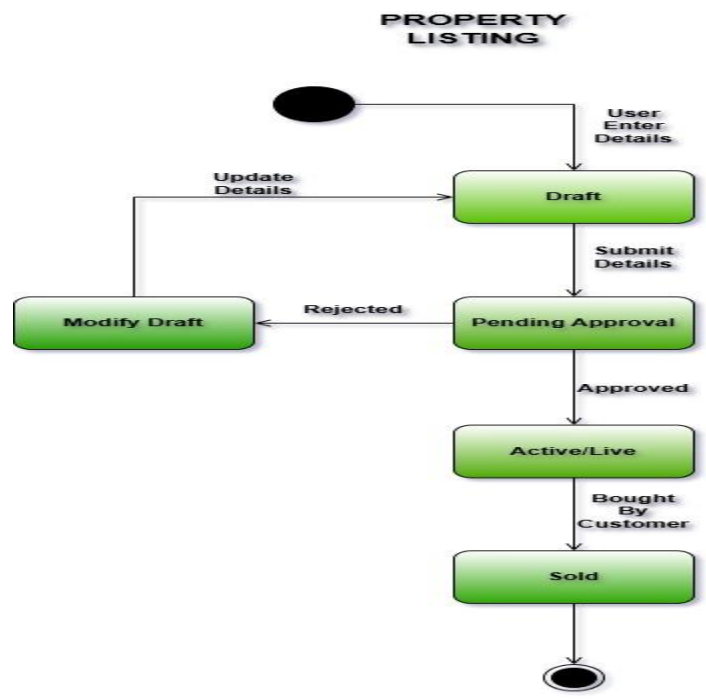
Conceptual Class Model



Activity Diagram



State Transition Diagrams (STDs)

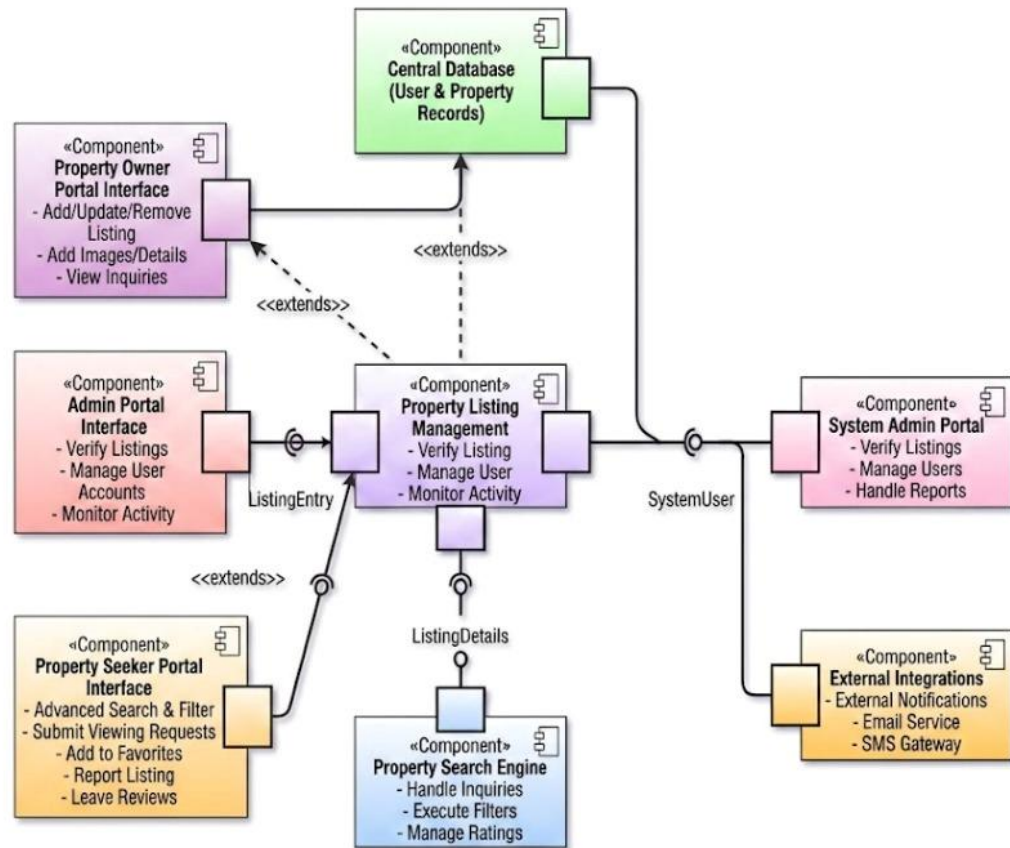


BOOKING / OFFER

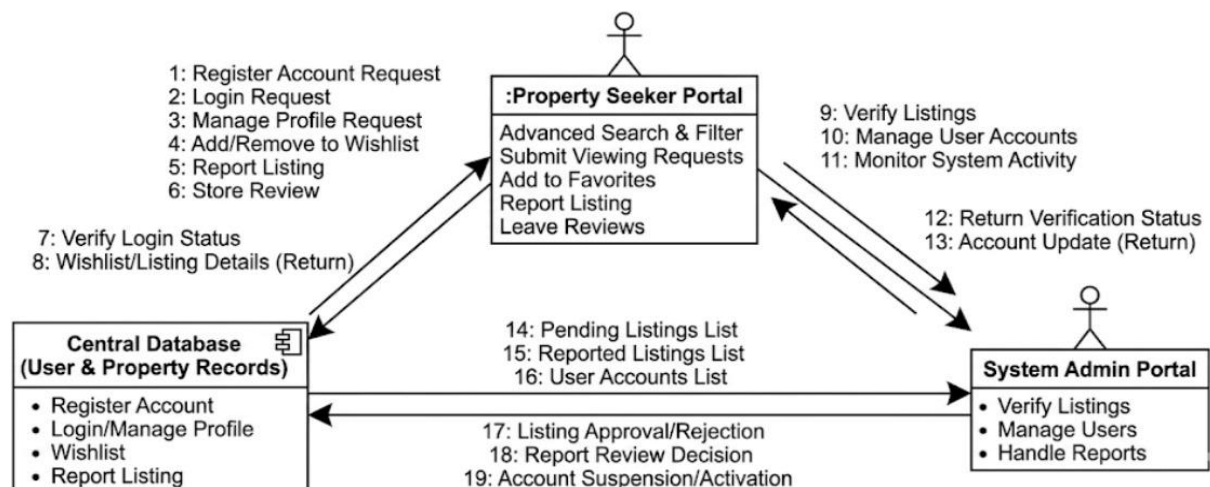


Component Based Diagram

PROPERTY SERVICES MANAGEMENT SYSTEM



Collaboration Diagram



These diagrams provide a complete representation of system behavior, structure, workflow, and data movement.

9. Future Enhancements

Future versions of Yali.com may include:

- Online Payment Integration
- AI-Based Property Recommendations
- Live Chat Support
- Mobile Application
- Google Maps Integration
- Property Verification Services
- Virtual Property Tours
- Advanced Analytics Dashboard

10. Conclusion

Yali.com successfully provides a modern and efficient platform for online property management. The system enables property owners to advertise properties, allows users to search and explore real estate opportunities, and provides administrators with complete control over system operations.

Through a well-structured software architecture, user-friendly interface, and comprehensive management capabilities, Yali.com demonstrates the practical application of software design and architecture principles in solving real-world property management challenges.